Ruiqi Chen

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Publication: Google scholar

EDUCATION

Washington University in St. Louis 2021 - 2026

- Ph.D., Neurosciences (degree expected in 2026); Thesis advisor: ShiNung Ching & Todd Braver
- Thesis: *Understanding whole-brain dynamics through generative modeling*

Peking University 2017 - 2021

- B.S., Intelligence Science and Technology; Thesis advisor: Si Wu
- Undergraduate Thesis: *Spatiotemporal information processing with reservoir decision-making networks*

RESEARCH EXPERIENCE

Washington University in St. Louis 2021 - 2026

Graduate Research Assistant; Advisors: ShiNung Ching & Todd Braver & Janine Bijsterbosch

Project: Predicting cognitive individual differences from resting state and task fMRI joint models (Preprint)

- Constructed a reliable individualized nonlinear dynamical model for joint analysis of resting & task fMRI.
- Revealed a unified generative mechanism underlying whole-brain dynamics across resting state and task conditions.
- Identified association between task performance and task-triggered topological & geometrical changes in dynamics.

Project: Diffeomorphic vector field alignment for model comparison and analysis (Manuscript in press)

- Developed a mathematical-computational framework to compare and analyze dynamical models.
- Implemented an optimization algorithm to identify the best possible orbit-aligning transformations.
- Demonstrated the efficiency of the method on high-dimensional nonlinear models fit on empirical neural data.

Project: Attractor landscape of individual resting state brain dynamics (Publication)

- Analyzed attractor landscape of over a thousand models of resting state brain dynamics from hundreds of participants.
- Discovered anatomically reliable nontrivial attractors embedded in functional brain networks.
- Illustrated that the resting state brain dynamics were close to bifurcation.

Collaboration: Control input synthesis for high-dimensional nonlinear systems (Publication)

- Implemented an algorithm to synthesize constant control inputs for nonlinear two-point-boundary-value problems.
- Helped improve the algorithm to synthesize minimum-energy time-varying control for general nonlinear systems.

Collaboration: Attentional fluctuation encoded in whole-brain dynamics geometry (Publication)

- Created a toolbox to train whole-brain dynamics model modulated by cognitive inputs.

Collaboration: Improving test-retest reliability estimation with HBM and MVPA (Publication)

- Trained Hierarchical Bayesian Models (HBM) for parcel-level neural activation during Stroop task.

Collaboration: Improving brain-behavior association by combining resting state and task fMRI (Publication)

- Computed functional connectivity from both resting state and task fMRI data to improve behavioral predictions.

University College London Summer 2020

Summer Research Assistant (Remote); Advisor: Sven Bestmann; (Project link)

- Simulated planar and spherical cortical traveling waves of different speeds, with static or dynamic sources, in different frequency bands, and under different levels of Signal-to-Noise-Ratio

Tsinghua University Summer 2019
Summer Research Assistant; Advisor: Bo Hong; (Project link)
- Conducted EEG experiments on subjects during resting state and while listening to story/music
- Performed traditional and functional-connectivity-based microstates clustering and Markov chain analyses

Peking University 2019 - 2020
Undergraduate Research Assistant; Advisor: Huan Luo; (Project link)
- Simulated an auditory sequential working memory cueing task with a Recurrent Neural Network (RNN)
- Designed an auditory working memory EEG experiment with different kinds of mental manipulation during retention period and collected and analyzed data from 16 subjects

GRANTS AND AWARDS

The Cognitive, Computational, and Systems Neuroscience (CCSN) Pathway Traineeship 2023 - 2025
$30,000/year; McDonnell Center for Systems Neuroscience, Washington University in St. Louis
The Cognitive, Computational, and Systems Neuroscience (CCSN) Pathway Travel Award 2024
$1,000; McDonnell Center for Systems Neuroscience, Washington University in St. Louis
Peking University Undergraduate Research Grant 2019 - 2020
4000 RMB; Peking University

TEACHING EXPERIENCE

Teaching Assistant, *Animal Behavior* Fall 2022
Washington University in St. Louis

SKILLS

- **Languages:** Chinese (native), Cantonese (native), English (proficient)
- **Experimental:** Psychtoolbox behavioral experiment, EEG recording, EEG & fMRI preprocessing
- **Programming**: MATLAB, Python, R; Linux operation; remote computing (Docker/Slurm); git; basic webpage development (HTML/CSS/Javascript); neural network training (Pytorch); Bayesian modeling fitting (brms)
- **Analytical:** Nonlinear dynamical modeling of empirical data, nonlinear representation alignment, nonlinear dimensionality reduction, limit set analysis and control input synthesis for nonlinear dynamical models, bifurcation and sensitivity analysis, controllability analysis, Hierarchical Bayesian Modeling, MVPA, time-varying connectivity

PEER-REVIEWED PUBLICATIONS

Chen, R., Vedovati, G., Braver, T., & Ching, S. (in press). Comparing Dynamical Models Through Diffeomorphic Vector Field Alignment. *Neural Computation*.

Chen, R., Singh, M., Braver, T. S., & Ching, S. (2025). Dynamical models reveal anatomically reliable attractor landscapes embedded in resting-state brain networks. *Imaging Neuroscience*, *3*, imag_a_00442.

Song, H., **Chen, R.**, Botch, T. L., Braver, T. S., Rosenberg, M. D., Zacks, J. M., & Ching, S. (2026). Geometry of neural dynamics along the cortical attractor landscape reflects changes in attention. *Nature Communications*, *17*(1), 2673.

Tamekue, C., **Chen, R.**, & Ching, S. (2025). On the Control of Recurrent Neural Networks using Constant Inputs. *IEEE Transactions on Automatic Control*, 1–16.

Freund, M. C., **Chen, R.**, Chen, G., & Braver, T. S. (2025). Complementary benefits of multivariate and hierarchical models for identifying individual differences in cognitive control. *Imaging Neuroscience*, *3*, imag_a_00447.

Easley, T., **Chen, R.**, Hannon, K., Dutt, R., & Bijsterbosch, J. (2023). Population modeling with machine learning can enhance measures of mental health—Open-data replication. *Neuroimage: Reports*, *3*(2), 100163.

PREPRINTS AND MANUSCRIPTS

Chen, R., Song, H., Ching, S., & Braver, T. S. (2026). *Task-induced topological and geometrical changes in whole-brain dynamics predict cognitive individual differences* (p. 2026.04.19.719533). bioRxiv. https://doi.org/10.64898/2026.04.19.719533

Chen, R., Vedovati, G., Braver, T., & Ching, S. (2024). *DFORM: Diffeomorphic vector field alignment for assessing dynamics across learned models* (No. arXiv:2402.09735). arXiv. https://doi.org/10.48550/arXiv.2402.09735

Yu, Z., Tamekue, C., **Chen, R.**, & Ching, S. (2026). *Scalable iterative Gramian synthesis for control-affine systems* [Manuscript submitted for publication]. Department of Electrical and Systems Engineering, Washington University in St. Louis.

CONFERENCE PRESENTATIONS

Chen, R., Singh, M. F., Braver, T. S., & Ching, S. (2024, October 7). *Towards a unifying mechanistic analysis framework for brain dynamics across resting and task states* [Conference presentation]. Annual Meeting of the Society for Neuroscience 2024, Chicago, IL, United States.

Chen, R., Singh, M. F., Braver, T. S., & Ching, S. (2023, November 13). *Resting state networks embed anatomically reliable nonlinear dynamics* [Conference presentation]. Annual Meeting of the Society for Neuroscience 2023, Washington, DC, United States.

REFERENCES

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